MECWIN®

EMPOWERING FUTURE

DEFENCE & AEROSPACE PORTFOLIO

Advanced Motion, Actuation & Electrical Systems for Defence and Aerospace

From design & development to qualified serial production — an Indian motion-systems partner built around your application.

MAKE IN INDIA









BLDC | PMSM | AC Induction | Axial Flux | EESM | Actuation | EM Brakes | Wiring Harness

About Mecwin

An Indian Defence & Aerospace Motion Systems Development and Manufacturing Partner

Indian design & manufacturing company Indigenous design ownership, aligned with Make in India and indigenisation programmes.

Bengaluru manufacturing base

Stamping, winding, machining, magnet assembly, potting and end-of-line testing under one roof.

In-house R&D and engineering

Electromagnetic, mechanical, thermal and electronics/interface design teams.

Prototype to serial production

The team that designs the product industrialises it and supports serial supply.



Assembly & test line, Bengaluru

Motor design › Electromagnetic design › Mechanical design › Electronics & interface › Manufacturing › Assembly › Testing

Our Defence & Aerospace Focus

Airborne systems

Auxiliary motors, cooling and actuation for fixed-and rotary-wing aircraft

Ground defence platforms

Turret, cooling, hatch and auxiliary drives for armoured vehicles

Naval / marine

Auxiliary motors and actuation, where applicable

UAV / drone

Propulsion, lift, gimbal and servo motors for military and ISR drones

eVTOL

Distributed electric propulsion and flight-control actuation

eSTOL

High power-density propulsion for short take-off aircraft

Electronics cooling & auxiliary

Fan, blower and pump motors for defence electronics

Radar & communication

Antenna and radar positioning drives with brakes and feedback

Electro-mechanical actuation

Linear, rotary and flap actuators as integrated assemblies

Special-purpose motion

Custom motion systems built to a programme requirement

Electric Motor Technology Portfolio



BLDC

Compact, high-speed

Outrunner and inrunner designs for UAV propulsion, fans, pumps and servo duty.



PMSM

High power density & efficiency

Sinusoidal-drive machines for propulsion, traction and precision positioning.



AC Induction

Rugged, magnet-free

Proven for scavenge fans, cooling fans and auxiliary drives in harsh duty.

CONFIGURED TO YOUR REQUIREMENT

Voltage	Torque	Speed	Mounting	Shaft	Cooling	Feedback

Airborne & UAV Motor Solutions



VTOL propulsion motor



2807 BLDC/PMSM drone motor



FPV / small-UAV motor



Cooling fan motors

MOTOR FUNCTIONS WE ADDRESS

Propulsion	Lift	Starter-generator	Pump	Compressor / blower
Cooling fan	Gimbal	Servo / positioning	Landing & ground support	High-altitude / high-speed

TARGET PLATFORMS

- Tactical & MALE UAVs
- Loitering munitions
- FPV & ISR drones
- Helicopters
- Fighter & transport aircraft

INDICATIVE SPECIFICATION RANGE

Power per motor
50 W – 30 kW

Supply
6S–24S packs, 28 VDC, 270 VDC

Speed
Up to 30,000 rpm

Operating temperature
−55 °C to +70 °C

Cooling
Natural, forced-air, conduction, liquid

Feedback
Sensorless, Hall, encoder, resolver

Indicative envelope. Final ratings are designed to the customer specification and duty profile.

eVTOL & eSTOL Electrification





Mecwin VTOL motor

- High power-density propulsion
- Distributed electric propulsion
- Axial-flux motor development
- Redundant winding concepts
- Liquid or air cooling options
- Low-weight designs
- High torque-to-weight ratio
- Motor + brake + actuator + harness integration

DEVELOPMENT PARTNERSHIP — PROTOTYPE THROUGH SERIALISATION

1. Requirement & architecture
2. Prototype & bench test
3. Iteration & flight-test support
4. Qualification support
5. Serialisation

Motorised Linear & Flap Actuators



Linear actuator — parachute ejection



Developed for automatic, reliable parachute deployment during landing.

ONE INTEGRATED ASSEMBLY

Motor	BLDC / PMSM / DC
Gearbox	Planetary, spur, ball / lead screw
Brake	Fail-safe holding brake
Sensor	Position & limit feedback
Controller / interface	Drive, connector, harness

▼

Tested actuator — single part number

INDICATIVE SPECIFICATION RANGE

Stroke
10 – 300 mm (custom)

Force / torque
Up to 10 kN linear · up to 200 Nm rotary

Supply
24 / 28 VDC, 270 VDC

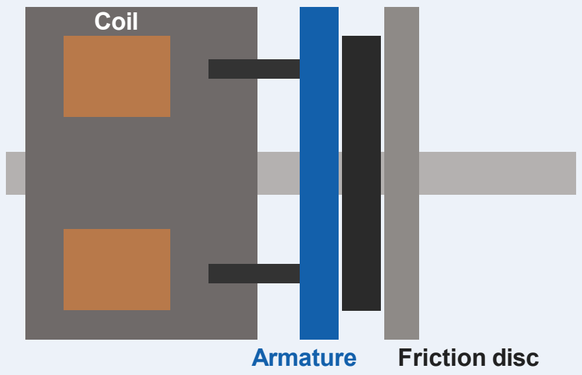
Feedback
Potentiometer, LVDT, Hall, absolute encoder

Stroke, load, speed and duty cycle set to the mission profile.

APPLICATIONS

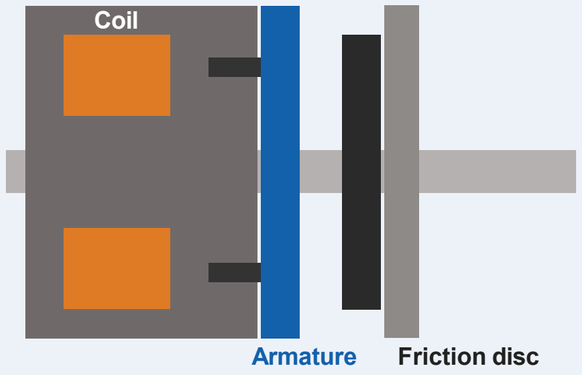
Flap actuation	**Door / hatch**	**Valve actuation**	**Antenna / radar positioning**
Payload positioning	**UAV control surfaces**	**Rotary actuators**	**Linear actuators**

Electromagnetic Brake Solutions



Power OFF — braked

Springs clamp the armature onto the friction disc. The load is held on loss of power (fail-safe).



Power ON — released

The coil is energised and pulls the armature back across the air gap. The shaft turns freely.

- **Fail-safe brakes**
- **Holding brakes**
- **Motor-integrated brakes**
- **Spring-applied, electrically released**
- **Compact customised brakes**
- **Brake-motor integrated units**

INDICATIVE RANGE

Static torque
0.5 – 100 Nm

Supply
24 / 28 VDC standard

Options
Manual release, monitor switch

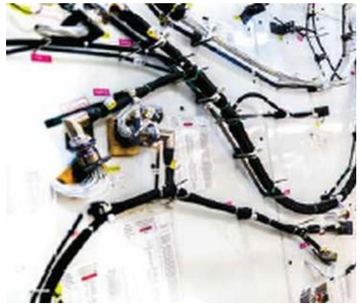
APPLICATIONS

- **Actuators**
- **Positioning systems**
- **Ground platforms**
- **UAV / eVTOL systems**
- **Special machinery**

Defence & Aerospace Wiring Harness



Platform harnesses



Aerospace harness boards



Hand-finished assemblies



RF cable assemblies

HARNESS TYPES

Power	Signal	Motor
Actuator	Sensor / feedback	Vehicle / platform

WHAT WE DELIVER

- Custom connectors & routing
- Shielding & EMI design
- Identification & traceability
- Build-to-print & build-to-spec
- Harness + motor / actuator integration

INDICATIVE SPECIFICATION RANGE

Conductor range
28 AWG – 4 AWG

Connectors
Mil-type circular, D-sub, RF, RJ45

Shielding
Overall and individual braid / foil

Testing
100% continuity, insulation resistance, hi-pot

Operating temperature
−45 °C to +85 °C

Materials, connectors and test levels follow the customer drawing or specification.

Custom Engineering & Development Capability

1 Requirement study
Specification, mission & duty profile

2 Concept
Topology & architecture trade-off

3 Electromagnetic design
FEA-based motor design

4 Mechanical design
3D models, drawings, tolerances

5 Thermal analysis
Thermal & CFD models

6 Prototype
In-house build, first article

7 Validation
Dyno, thrust stand, environmental

8 Qualification support
Customer qualification plan

9 Tooling & industrialisation
Dies, fixtures, process sheets

10 Serial production
Traceable, SAP-controlled supply

Reverse engineering
Where legally and customer authorised

Form-fit-function replacement
Drop-in replacement for existing equipment

Obsolescence & indigenisation
Indigenous source for imported or obsolete parts

DESIGN & SIMULATION TOOLCHAIN

Simcenter Motorsolve · Simcenter MAGNET · Ansys Motor-CAD · MATLAB Simulink · KiCad · ptc creo · SOLIDWORKS · Ansys CFX · CFD · CFturbo · SAP

ERP-controlled production & traceability

Qualification & Defence-Grade Engineering

IN-HOUSE TEST CAPABILITY



Salt spray

Salt mist / corrosion



Dust ingress

IP protection



Thermal chamber

Temperature & humidity



Vibration shaker

With climatic chamber



Motor test dyno

Performance mapping



Dynamic balancing

Rotor balance



Life-cycle test rig

Endurance testing



Thrust stand

Motor + propeller data

Products are designed to customer qualification requirements. Formal qualification status is specific to each product and programme.

QUALIFICATION SUPPORT

Planned with the customer; executed in-house or at accredited laboratories

- Shock, altitude and extended environmental tests
- EMI / EMC tests
- MIL-STD-810, MIL-STD-461, RTCA DO-160, IEC / IS — as the customer specifies
- Customer-specific qualification plans and reports

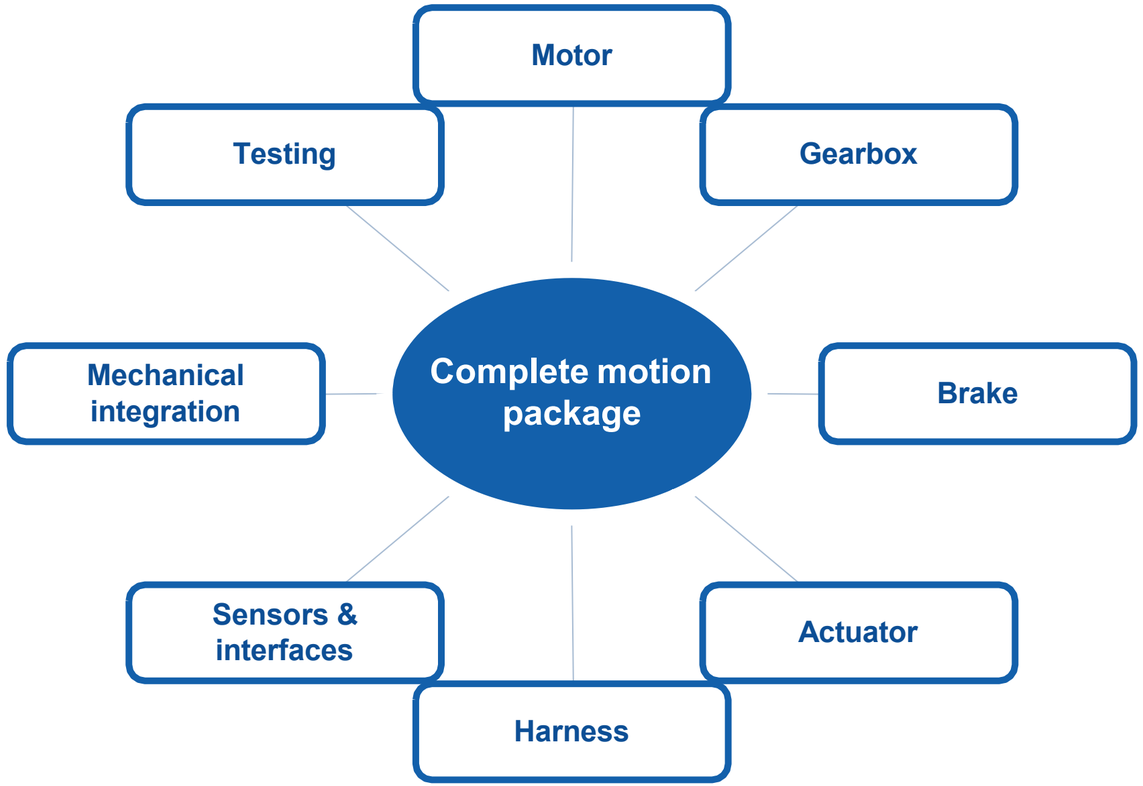
ENGINEERED-IN ROBUSTNESS

Insulation, VPI & potting	**Special coatings**
IP sealing	**EMI / EMC design**
Traceability	**Documentation**

Turnkey Project Capability

Application requirement → Engineering → Motor · Actuator · Brake · Harness → Qualification → Integration → Serial production

	Component-only supplier	Mecwin turnkey
Scope	Single component to drawing	**Complete motion package**
Interfaces	Customer integrates and debugs	**Mecwin owns mechanical & electrical interfaces**
Testing	Part-level inspection	**System-level test before delivery**
Accountability	Split across several vendors	**One project owner**
Supply	Multiple purchase orders	**One part number, serial supply**



HOW WE ENGAGE

Our Engagement & Manufacturing Model

PREFERRED

Serial manufacturing

Long-term production of Mecwin-developed or co-developed products against a firm demand schedule.

Capability

WELCOMED

Strategic co-development

Co-development and indigenisation projects with clear production potential.

Capability

ACCEPTED

Turnkey projects

Complete motion packages where Mecwin has meaningful engineering and manufacturing scope.

Capability

SELECTIVE

Build-to-print / subcontract

Only where volumes justify capacity allocation, with a minimum one-year sustainable production load.

Capability

MANUFACTURING BASE, BENGALURU



High-speed press line



Automatic coil winding



Magnet assembly



CNC machining



End-of-line motor testing



Gear hobbing



Mecwin Group Contact

Mecwin Technologies India Pvt. Ltd.
Customer Care: **+91 97412 29797**
Email: marketing@mecwinindia.com

Mecwin Technologies India Pvt. Ltd.
Ph: **+44 74040 31886**
Email: sales@mecwinindia.com
12 Amber Close, Rugby CV22 5GE, United Kingdom

Mecwin Infinity Tech Park, #42/2,
Opp. Machohalli Lake, K.G. Lakkenahalli Road,
Machohalli Village, Bengaluru North Taluk-560091, India

LinkedIn YouTube Instagram